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Balance assistance and fireproofing device for scooters

Abstract

Disclosed herein are exemplary devices that provide balance assistance and fireproofing to electric scooters. The devices described herein can be clipped onto existing scooters, providing balance improvements via one or more training wheels and providing fireproofing improvements via one or more units containing fire-suppressant materials. A device may be attached to the center of a scooter's standing platform to improve the stability of the scooter and to provide fireproofing to the batteries. The device can include two training wheels, each positioned to an opposite side of the standing platform, to provide the vehicle with stability. The training wheels may be connected to an axle that is mounted to a fireproofing unit. The fireproofing unit, which contains fire-suppressant materials, can clamp to the standing platform, which contains at least a portion of the batteries. Users can adjust, reposition, and/or remove portions of the balance assistance and fireproofing device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/577,402, filed Apr. 21, 2023, the disclosure of which is incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to scooters, and more specifically, to a balance assistance and fireproofing device compatible with scooters.

BACKGROUND OF THE DISCLOSURE

(2) Personal scooters, such as electric scooters, are commonly used for local transportation. Current self-balancing electric scooter designs typically include a standing platform for a user, two in-line wheels beneath the standing platform, motors and batteries to drive the wheels, and handlebars to control the front wheel. One of the problems with two-wheeled, self-balancing scooters, both non-electric and electric, is that they can be difficult to ride and/or stand on without falling, especially for elderly, disabled, or untrained individuals. Even with proper guidance and training, the use of scooters can feel quite unstable and difficult to use, and may result in injury or even death to the user. When operating a scooter, users typically stand on the standing platform with one foot in front and another foot behind, or suspended in the air, due to the limited surface area of the standing platform. Due to the design of the scooter and the standing position of the user, many users may find it difficult to balance on scooters without tipping over and/or falling. This increases the risk of injury for the user.

(3) Furthermore, another problem with current electric scooter designs is that the batteries are prone to combustion. The batteries typically are lithium-based or nickel-based batteries that may occasionally combust or generate smoke, especially while charging or during operation of the electric scooters. This increases the risk of injury to the user and fire damage to the surroundings of the electric scooters. Instead of returning and/or retiring the electric scooters, which may be wasteful and/or expensive, users may wish to modify existing electric scooters to mitigate the risk of injury and fire damage.

SUMMARY OF THE DISCLOSURE

(4) As stated above, current electric scooters fail to implement balancing and fireproofing features, and, as a result, users of such electric scooters may experience increased risk of injury and fire damage. Furthermore, the users may wish to modify existing electric scooters to implement these features to reduce waste and expenses. Thus, Applicant discovered an improved accessory for a scooter, that can provide users with balancing and fireproofing features and can be retrofittable onto existing scooters. As described in examples herein, a balance assistance and fireproofing device for scooters can provide these improvements. The balance assistance and fireproofing device can be attached or clipped onto existing scooters, providing balance improvements via one or more training wheels and providing fireproofing improvements via one or more units containing fire-suppressant materials. Thus, the balance assistance and fireproofing device can improve the safety of users in an easy-to-install, adjustable, and/or inexpensive manner. Accordingly, disclosed herein are devices that may address some of the above-identified needs.

(5) In some embodiments, a balance assistance and fireproofing device may be attached to the center of an electric scooter's standing platform, to improve the stability of the scooter and to provide fireproofing to the batteries. The attachment can include two training wheels, each positioned to an opposite side of the standing platform, to provide the vehicle with stability. The training wheels may be connected to an axle that is mounted to a fireproofing unit. The fireproofing unit, which contains fire-suppressant materials, can clamp to the standing platform, which contains at least a portion of the batteries. Users can easily adjust, reposition, and/or remove portions of the balance assistance and fireproofing device. For example, after improving their expertise in balancing and riding the scooter, the users can remove the training wheels, while leaving the fireproofing unit in place. Alternatively, in another example, users can remove the entire balance

assistance and fireproofing device from one scooter and attach it to another scooter. In some embodiments, the training wheels can be adjusted and/or replaced in order to mount the balance assistance and fireproofing device to different scooters. Moreover, each of the training wheels can adjust to the base dimensions of the scooter in regards to the length and depth of the standing platform. In some embodiments, the distance between the training wheels and the support platform of the scooter can also be adjusted. The case of adjustability and safety features are among some of the advantages of the balance assistance and fireproofing devices disclosed herein.

(6) According to some embodiments, an exemplary device configured to be attached to a scooter comprises: a top layer attached to a bottom layer such that the top and bottom layers surround at least a portion of a standing platform of the scooter when attached to the scooter, wherein the top layer and/or bottom layer comprise a fire-suppressant material; a rod attached to the top or bottom layer; and two wheels attached to the rod such that the two wheels are situated on opposite sides of the attached top and bottom layers.

(7) In some embodiments, the top layer comprises one or more grooves that engage with one or more corresponding protrusions of the bottom layer when the top layer is attached to the bottom layer. In some embodiments, the top layer and the bottom layer are configured to slide together such that the one or more grooves of the top layer engage with the one or more protrusions of the bottom layer to surround at least a portion of the standing platform of the scooter.

(8) In some embodiments, the top layer comprises one or more protrusions that engage with one or more corresponding grooves of the bottom layer when the top layer is attached to the bottom layer. In some embodiments, the top layer and/or bottom layer comprises a groove configured to receive the standing platform.

(9) In some embodiments, the top layer and/or bottom layer comprises a storage chamber.

(10) In some embodiments, the fire-suppressant material comprises a fireproof sealant comprising one or more of: caulk, foam, silicone, and putty.

(11) In some embodiments, the top or bottom layer comprises one or more brackets and the rod is attached to the one or more brackets. In some embodiments, the rod is attached to a bottom surface of the bottom layer.

(12) In some embodiments, the top layer and/or bottom layer extends laterally outward from the standing platform of the scooter when attached to the scooter.

(13) In some embodiments, the standing platform comprises a battery and at least a portion of the standing platform comprising the battery is surrounded by the top and bottom layers.

(14) According to some embodiments, an exemplary scooter system comprises: a scooter comprising a standing platform and a battery at least partially embedded within the standing platform; and a device configured to be attached to the scooter, the device comprising: a top layer attached to a bottom layer such that the top and bottom layers surround at least a portion of the standing platform of the scooter when attached to the scooter, wherein the top layer and/or bottom layer comprise a fire-suppressant material; a rod attached to the top or bottom layer; and two wheels attached to the rod such that the two wheels are situated on opposite sides of the attached top and bottom layers.

(15) It will be appreciated that any of the variations, aspects, features and options described in view of the systems can be combined.

(16) Additional advantages will be readily apparent to those skilled in the art from the following detailed description. The aspects and descriptions herein are to be regarded as illustrative in nature and not restrictive.

(17) All publications, including patent documents, scientific articles and databases, referred to in this application are incorporated by reference in their entirety for all purposes to the same extent as if each individual publication were individually incorporated by reference. If a definition set forth herein is contrary to or otherwise inconsistent with a definition set forth in the patents, applications,

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) Exemplary embodiments are described with reference to the accompanying figures, in which:
- (2) FIG. 1A depicts an overall view of an exemplary balance assistance and fireproofing device compatible with a scooter, in accordance with some embodiments disclosed herein.
- (3) FIG. 1B depicts an upper perspective view of an exemplary balance assistance and fireproofing device compatible with a scooter, in accordance with some embodiments disclosed herein.
- (4) FIG. 1C depicts a lower perspective view of an exemplary balance assistance and fireproofing device compatible with a scooter, in accordance with some embodiments disclosed herein.
- (5) FIG. 2A depicts an exemplary wheel of a balance assistance and fireproofing device, in accordance with some embodiments disclosed herein.
- (6) FIG. 2B depicts an exemplary wheel rim of a balance assistance and fireproofing device, in accordance with some embodiments disclosed herein.
- (7) FIG. 3A depicts a bottom perspective view of an exemplary top layer of a fireproofing central unit of a balance assistance and fireproofing device, in accordance with some embodiments disclosed herein.
- (8) FIG. 3B depicts a top perspective view of an exemplary top layer of a fireproofing central unit of a balance assistance and fireproofing device, in accordance with some embodiments disclosed herein.
- (9) FIG. 4 depicts an exemplary bottom layer of a fireproofing central unit of a balance assistance and fireproofing device, in accordance with some embodiments disclosed herein.
- (10) FIG. 5 depicts exemplary grooves and protrusions of a fireproofing central unit of a balance assistance and fireproofing device, in accordance with some embodiments disclosed herein.
- (11) FIG. 6 depicts an exemplary axle of a balance assistance and fireproofing device, in accordance with some embodiments disclosed herein.
- (12) FIG. 7 depicts an exemplary bracket of a balance assistance and fireproofing device, in accordance with some embodiments disclosed herein.

DETAILED DESCRIPTION OF THE DISCLOSURE

- (13) Disclosed herein are exemplary scooter accessories that provide balance assistance and fireproofing to scooters. The balance assistance and fireproofing devices described herein can be attached or clipped onto existing scooters, providing balance improvements via one or more training wheels and providing fireproofing improvements via one or more units containing fire-suppressant materials. In some embodiments, a balance assistance and fireproofing device may be attached to the center of a scooter's standing platform, to improve the stability of the scooter and to provide fireproofing to the batteries. The attachment can include two training wheels, each positioned to an opposite side of the standing platform, to provide the vehicle with stability. The training wheels may be connected to an axle that is mounted to a fireproofing unit. The fireproofing unit, which contains fire-suppressant materials, can attach or clamp to the standing platform, which contains at least a portion of the batteries. Users can easily adjust, reposition, and/or remove portions of the balance assistance and fireproofing device.
- (14) As described above, current scooters fail to implement balancing and/or fireproofing features, and, as a result, users of such scooters may experience increased risk of injury and fire damage. For example, conventional two-wheeled, self-balancing scooters can be difficult to ride and/or stand on without falling, especially for elderly, disabled, or untrained individuals, and the batteries of such conventional scooters may be prone to combustion. Conversely, the devices described herein can

provide users with balancing and fireproofing features and are retrofittable onto existing scooters. In some embodiments, a balance assistance and fireproofing device can be clipped onto existing scooters, providing balance improvements via one or more training wheels and providing fireproofing improvements via one or more units containing fire-suppressant materials. Thus, the balance assistance and fireproofing device can improve the safety of users in an easy-to-install, adjustable, and/or inexpensive manner.

(15) Reference will now be made in detail to implementations and embodiments of various aspects and variations of systems, devices, and methods described herein. Although several exemplary variations of the systems, devices, and methods are described herein, other variations of the systems, devices, and methods may include aspects of the systems, devices, and methods described herein combined in any suitable manner having combinations of all or some of the aspects described. Disclosed herein are devices that may address one or more of the problems discussed above.

(16) FIG. 1A depicts an overall view of an exemplary balance assistance and fireproofing device **100** compatible with a scooter **198**, in accordance with some embodiments. The scooter **198** can be any scooter on the market, such as a self-balancing scooter with two inline wheels. In some embodiments, the scooter **198** comprises a standing platform **199** for a user, two in-line wheels beneath the standing platform, motors and batteries to drive the wheels, and handlebars to control the front wheel. It is to be understood, however, that the scooter **198** can have additional configurations and is not limited to this combination of features.

(17) The standing platform **199** for the user can include an approximately flat surface that provides a user with a place to stand during operation of the scooter **198**. In some embodiments, the balance assistance and fireproofing device **100** is configured to attach to at least a portion of the standing platform **199** to provide balancing and fireproofing functionality to the scooter **198**. The device **100** can attach to the standing platform **199** along any region or the entirety of the standing platform **199**. For example, the device **100** can attach to a portion of the standing platform **199** near the rear in-line wheel of the scooter **198**, near the front in-line wheel of the scooter **198**, centered between the two in-line wheels of the scooter **198**, or anywhere in-between. In some embodiments, the device **100** can attach to the entire length of the standing platform **199**.

(18) In some embodiments, the device **100** can include multiple components, including a fireproofing central unit comprising a top layer **104T** and a bottom layer **104B**, a clamp mechanism **106** for attaching the central unit to the scooter **198**, a rod **108** attached to the central unit, and two wheels **102L** and **102R** attached to the rod **108**. The device **100** is described in greater detail in the examples below.

(19) FIG. 1B depicts an upper perspective view of an exemplary balance assistance and fireproofing device **100** compatible with a scooter **198**, in accordance with some embodiments. In some embodiments, the device **100** can include a fireproofing central unit comprising a top layer **104T** and a bottom layer **104B**. The central unit can include a clamp mechanism **106** for removably attaching the central unit to the scooter deck **199** of the scooter **198**. Beneath the standing platform **199**, a rod **108** (not pictured) runs underneath and is connected to the bottom layer **104B** of the central unit. The rod **108** can serve as an axle that is connected to two wheels, a left wheel **102L** and a right wheel **102R**. Taken together, these components of the device **100** can provide balance assistance and fireproofing functionalities when attached to the scooter **198**.

(20) In some embodiments, the training wheels **102L** and **102R** can be any tire, wheel, roller, and/or similar component. Each wheel can be situated on a diametrically opposite side of the scooter **198** from the other wheel (e.g., left or right). During operation of a scooter **198** with the device **100** attached, the training wheels **102L** and **102R** can be configured to make contact with the ground at least part of the time. The training wheels **102L** and **102R** can prevent the scooter **198** from tipping over to the left and/or right, thereby providing balance assistance to the user while operating the scooter **198**. For example, if the user leans left, the training wheel **102L** may make

contact with the ground, preventing the scooter **198** from tipping further left. In some embodiments, the training wheels **102L** and **102R** may be the same size as the wheels of the scooter **198**. It is to be understood, however, that the training wheels **102L** and **102R** can be of any size, width, diameter, and/or form factor. In some embodiments, the distance between the training wheels **102L** and **102R** and the sides of the standing platform **199** can be adjustable. In some embodiments, the training wheels **102L** and **102R** can be removed from the device **100**. In such embodiments, the device **100** may operate without the balance assistance features and may instead function as a fireproofing device.

(21) FIG. 2A depicts a close-up view of an exemplary wheel **202** of a balance assistance and fireproofing device (e.g., device **100** of FIG. 1), in accordance with some embodiments. The outer portion of the wheel **200** may be any shock-absorption wheel, such as a rubber wheel. FIG. 2B depicts a close-up view of an exemplary wheel rim **202x** of the wheel **200**. The inner portion of the wheel **200** can include any wheel rim **202x** compatible with the outer portion of the wheel **200**. The wheel rim **202x** can connect the wheel **202** to the axle (e.g., rod **108** of FIG. 1). The wheel **202** can share any characteristics of the training wheels **102L** and **102R** of FIG. 1, and vice versa.

(22) Referring back to FIG. 1B, in some embodiments, the fireproofing central unit can be a structure consisting at least partially of a fire-suppressant material (e.g., a fireproof sealant comprising one or more of: caulk, foam, silicone, and putty). The fire-suppressant materials can line at least a portion of the interior of the central unit. In some embodiments, the fire-suppressant materials can coat at least a portion of the exterior of the central unit.

(23) In some embodiments, the central unit can be a box-like structure that removably attaches to, surrounds, and/or encloses at least a portion of the standing platform **199**. The central unit can comprise a top layer **104T** and a bottom layer **104B** which can be connected by a clamp mechanism **106**. The top layer **104T** and bottom layer **104B**, when clamped together around the standing platform **199**, can enclose a battery embedded at least partially within the standing platform **199**. This can enable the central unit, which contains fire-suppressant materials, to prevent battery fires from damaging the user and/or the surroundings of the scooter **198**. In some embodiments, the top layer **104T** and bottom layer **104B**, when clamped together around the standing platform **199**, can enclose and/or surround at least a portion of the top, bottom, left, and right sides of the standing platform **199**. In some embodiments, the top surface of the bottom layer **104B** can engage with and/or attach to the bottom surface of the top layer **104T**.

(24) In some embodiments, the top layer **104T** can comprise a solid surface upon which the user can stand. In some embodiments, the top layer **104T** can contain fire-suppressant materials. As shown in FIG. 1B, the surface of the top layer **104T** may be textured to provide a user with improved traction and grip relative to a smooth surface. The top layer **104T** can be positioned over a top surface of the standing platform **199**. In some embodiments, the bottom surface of the top layer **104T** can attach to the top surface of the bottom layer **104B**. For example, the top layer **104T** can attach to the bottom layer **104B** such that the layers surround at least a portion of a standing platform **199** of the scooter **198** when attached to the scooter **198**.

(25) In some embodiments, a surface of the top layer **104T** may be elevated above the surface of the standing platform **199**. In some such embodiments, the top layer **104T** can extend vertically upward from the standing platform **199** when attached to the scooter **198**. In some embodiments, the top layer **104T** of the central unit extends laterally (also referred to herein as “horizontally”) outward from the standing platform **199** when attached to the scooter **198**. This can provide a larger surface upon which the user can stand. In some embodiments, a user can place both feet side-by-side on the surface of the top layer **104T**. Standing with feet side-by-side can improve user comfort and improve the user's balance, relative to standing with one foot in front of the other on the standing platform **199**.

(26) FIGS. 3A and 3B depict close-up views of an exemplary top layer **304T** of a central fireproofing unit of a balance assistance and fireproofing device (e.g., device **100** of FIG. 1), in

accordance with some embodiments. In some embodiments, the top layer **304T** can include one or more components for attaching the top layer **304T** to a bottom layer (e.g., bottom layer **404B** of FIG. 4). In some embodiments, the top layer **304T** can include one or more components for receiving and/or attaching the top layer **304T** to the standing platform of a scooter. The top layer **304T** can share any characteristics of the top layer **104T** of FIG. 1, and vice versa.

(27) FIG. 3A depicts a bottom perspective view of the top layer **304T**. Specifically, FIG. 3A illustrates a first surface **304Ta** of the top layer **304T**. As shown in FIG. 3A, the first surface **304Ta** can be a bottom surface of the top layer **304T**. However, it is to be understood that the first surface **304Ta** can be any surface of the top layer **304T**, such as a top surface or a side surface, and is not limited to what is illustrated in FIG. 3A.

(28) In some embodiments, the top layer **304T** can include one or more grooves and/or one or more protrusions that can engage with one or more corresponding protrusions and/or one or more grooves on a bottom layer (e.g., bottom layer **404B** of FIG. 4). For example, in some embodiments, the first surface **304Ta** includes one or more grooves that engage with one or more corresponding protrusions of the bottom layer when the top layer **304T** is attached to the bottom layer. In some embodiments, the first surface **304Ta** includes one or more protrusions that engage with one or more corresponding grooves of the bottom layer when the top layer **304T** is attached to the bottom layer.

(29) In some embodiments, a groove can be any physical indentation, furrow, channel, or receptacle on the surface of an object. In some embodiments, a groove on the top layer **304T** can be an indentation that runs at least partially along any dimension (length, width, depth) of the top layer **304T**. For example, as shown in FIG. 3A, the grooves of the top layer **304T** can run fully along one dimension of the first surface **304Ta** of the top layer **304T**. The grooves can have any shape and/or cross-section. For example, the groove can have a circular, triangular, rectangular, trapezoidal, sawtooth, or frustrum-shaped cross-section. Further, each groove can vary in size and/or shape throughout the length of the groove.

(30) In some embodiments, a protrusion can be any physical projection, ridge, or extrusion on the surface of an object. In some embodiments, a protrusion on the top layer **304T** can be a ridge that runs at least partially along any dimension (length, width, depth) of the top layer **304T**. In some embodiments, the protrusions of the top layer **304T** can run fully along one dimension of the first surface **304Ta** of the top layer **304T**.

(31) In some embodiments, a groove can be configured to receive a corresponding protrusion, and a protrusion can be configured to engage with a corresponding groove. The shapes of the protrusion and groove can be chosen to enable at least a portion of the groove and at least a portion of the protrusion to be positioned flush against one another. In some embodiments, the shapes of the protrusion and groove can be inverse shapes. For example, both the protrusion and the groove can have a square cross-section of approximately equal dimensions.

(32) In some embodiments, a groove can be configured to receive the standing platform of the scooter. The shape of the groove can be chosen to enable at least a portion of the groove and at least a portion of the standing platform to be positioned flush against one another. For example, the groove can be chosen to match the dimensions of the standing platform such that, when the top layer **304Ta** is placed on top of the standing platform, the groove is positioned flush against the top of the standing platform. In some embodiments, the groove can be configured to engage with the standing platform to reduce and/or eliminate the unintentional movement and/or slipping of the top layer **304T** against the standing platform. In some embodiments, the groove can include a friction-based anti-slip mechanism, such as Velcro or nonslip padding, to engage with the standing platform. In some embodiments, the groove can include a buckle, lock, strap, latch, or other connective mechanism configured to engage with the standing platform.

(33) FIG. 3B depicts a top perspective view the top layer **304T**. Specifically, FIG. 3B illustrates a second surface **304Tb** of the top layer **304T**. As shown in FIG. 3B, the second surface **304Tb** can

be a top surface of the top layer **304T**. However, it is to be understood that the second surface **304Tb** can be any surface of the top layer **304T**, such as a bottom surface or a side surface, and is not limited to what is illustrated in FIG. **3B**. In some embodiments, at least a portion of the second surface **304Tb** can include a non-slip coating, material, and/or texture. The second surface **304Tb** can be configured to provide a user with a solid surface upon which to stand when operating the scooter.

(34) In some embodiments, the top layer **304T** can be at least partially hollow. In some embodiments, the top layer **304T** can contain a hollow storage container for storing items. The storage container can be a lockable compartment secured within the hollow regions of the top layer **304T**.

(35) Referring back to FIG. **1B**, in some embodiments, the bottom layer **104B** can comprise a structure containing fire-suppressant materials. The bottom layer **104B** can be positioned over a bottom surface of the standing platform **199**. In some embodiments, the top surface of the bottom layer **104B** can attach to the bottom surface of the top layer **104T**. In some embodiments, a rod **108** can attach to the central unit via one or more fixed points on a bottom surface of the bottom layer **104B**.

(36) In some embodiments, the bottom layer **104B** can be at least partially hollow. In some embodiments, the bottom layer **104B** can contain a hollow storage container for storing items. The storage container can be a lockable compartment secured within the hollow regions of the bottom layer **104B**.

(37) FIG. **4** depicts a close-up view of an exemplary bottom layer **404B** of a central fireproofing unit of a balance assistance and fireproofing device (e.g., device **100** of FIG. **1**), in accordance with some embodiments. In some embodiments, the bottom layer **404B** can include one or more components for attaching the bottom layer **404B** to a top layer (e.g., top layer **304T** of FIGS. **3A** and **3B**). For example, the bottom layer **404B** can include one or more grooves and/or one or more protrusions that can engage with one or more corresponding protrusions and/or one or more grooves on a top layer. The bottom layer **404B** can share any characteristics of the bottom layer **104B** of FIG. **1**, and vice versa.

(38) In some embodiments, the bottom layer **404B** includes a first surface **404Ba**. As shown in FIG. **4**, the first surface **404Ba** can be a top surface of the bottom layer **404B**. However, it is to be understood that the first surface **404Ba** can be any surface of the bottom layer **404B**, such as a bottom surface or a side surface, and is not limited to what is illustrated in FIG. **4**.

(39) In some embodiments, the bottom layer **404B** can include one or more grooves and/or one or more protrusions that can engage with one or more corresponding protrusions and/or one or more grooves on a bottom layer (e.g., top layer **304T** of FIGS. **3A** and **3B**). For example, in some embodiments, the first surface **404Ba** includes one or more grooves that engage with one or more corresponding protrusions of the bottom layer when the bottom layer **404B** is attached to the top layer. In some embodiments, the first surface **404Ba** includes one or more protrusions that engage with one or more corresponding grooves of the bottom layer when the bottom layer **404B** is attached to the top layer. The grooves and protrusions of the bottom layer **404B** can share any characteristics of the grooves and protrusions of FIGS. **3A** and **3B**, and vice versa.

(40) In some embodiments, the bottom layer **404B** includes a second surface **404Bb**. As shown in FIG. **4**, the second surface **404Bb** can be a bottom surface of the bottom layer **404B**. However, it is to be understood that the second surface **404Bb** can be any surface of the bottom layer **404B**, such as a top surface or a side surface, and is not limited to what is illustrated in FIG. **4**. In some embodiments (not pictured), the second surface **404Bb** can include one or more attachment points and/or fixed points for attaching a rod (e.g., rod **108** of FIG. **1**) to the bottom layer **404B**.

(41) Referring back to FIG. **1B**, in some embodiments, at least one clamp mechanism **106** can hold the top layer **104T** and bottom layer **104T** together around the standing platform **199**. The clamp mechanism **106** can function as a means for attaching the central unit to the standing platform **199**,

and by extension, the scooter **198**. The clamp mechanism **106** can be affixed to the central unit and situated between the top layer **104T** and the bottom layer **104B** of the central unit. In some embodiments, the clamp mechanism **106** comprises a top segment (e.g., attached to top layer **104T**) and a bottom segment (e.g., attached to bottom layer **104B**). The segments can be positioned on diametrically opposite sides of the standing platform **199** (e.g., top and bottom sides). The top segment and the bottom segment can be engageable with one another to define the clamp mechanism **106**.

(42) In some embodiments, the clamp mechanism **106** can comprise a groove-and-protrusion attachment. The grooves and protrusions of the clamp mechanism **106** can have any characteristics of the grooves and protrusions described in FIGS. 3A, 3B, and 4, and vice versa.

(43) As shown in FIG. 1B, the clamp mechanism **106** can comprise the grooves and protrusions along the junction of the top layer **104T** and bottom layer **104B**. The grooves and protrusions can form zig-zagging surfaces along at least a portion of the top layer **104T** and at least a portion of the bottom layer **104B**. The grooves may be configured to receive the protrusions (e.g., the shape of the grooves and protrusions are inverse) and to lock the protrusions in place. The width of the grooves and protrusions may be variable to enable such locking. In some embodiments, a groove **106Tg** on the top layer **104T** can engage with a protrusion **106Bp** on the bottom layer **104B**. The groove **106Tg** on the top layer **104T** comprises the top segment of the clamp mechanism **106**, and the protrusion **106Bp** on the bottom layer **104B** comprises the bottom segment of the clamp mechanism **106**. By slotting and/or sliding the protrusion **106Bp** into the groove **106Tg**, the clamp mechanism **106** can hold the top layer **104T** and the bottom layer **104B** together around the standing platform **199**. For example, the top layer **104T** can be placed on a top side of the standing platform **199** such that the groove **106Tg** is on a bottom-facing surface of the top layer **104T**. The bottom layer **104B** can be placed on a bottom side of the standing platform **199** such that the protrusion **106Bp** is on a top-facing surface of the bottom layer **104B**. The top layer **104T** and bottom layer **104B** can slide laterally toward one another such that they are flush with one another. The protrusion **106Bp** can slide into the groove **106Tg**, such that the top layer **104T** and bottom layer **104B** cannot be separated by vertical forces (e.g., gravity). The union of the groove **106Tg** and protrusion **106Bp** can thus hold the top layer **104T** and bottom layer **104B** together around and/or surrounding the standing platform **199**.

(44) In some embodiments, the device **100** can include multiple clamp mechanisms **106**. For example, FIG. 5 illustrates an exemplary fireproofing central unit comprising two clamp mechanisms **506**. The top layer **504T** contains two grooves **506Tg**, and the bottom layer **504B** contains two protrusions **506Bp**. The grooves and protrusions are located on either side (e.g., left and right) of a standing platform **599**. The top layer **504T** and bottom layer **504B** can slide laterally toward one another such that they are flush with one another. The protrusions **506Bp** can slide into the grooves **506Tg**, such that the top layer **504T** and bottom layer **504B** cannot be separated by vertical forces (e.g., gravity). The union of the multiple grooves **506Tg** and multiple protrusions **506Bp** can thus hold the top layer **504T** and bottom layer **504B** together around the standing platform **599**.

(45) In some embodiments, a groove **507** adjacent to the clamp mechanism **106** can be engageable with a portion of the standing platform **199** of the scooter **198**. Engaging with the standing platform can include being removably affixed to a portion of the standing platform **199** such that the clamp mechanism **106**, and by extension the device **100**, does not slide back and forth along the standing platform **199** while the scooter **198** is in operation. The groove **507** can engage with the portion of the standing platform **199** enclosing at least a portion of a battery **508** at least partially embedded within the standing platform **199**. This can position the fireproofing central unit around the battery **508**, containing it and preventing potential battery fires from spreading to the user or the surroundings of the scooter. It is to be understood that, although the groove **507** is illustrated as being on the top layer **504T** in FIG. 5, the groove **507** can be on any layer, including the bottom

layer **504B**. In some embodiments, both the top layer **504T** and the bottom layer **504B** can include a groove **507** for engaging with the standing platform **199**. The groove **507** can have any characteristics of the grooves and protrusions described in FIGS. **3A**, **3B**, and **4**, and vice versa.

(46) Referring back to FIG. **1B**, the clamp mechanism **106** of FIG. **1** can share any characteristics of the clamp mechanisms **506** of FIG. **5**, and vice versa.

(47) In some embodiments (not pictured), the clamp mechanism **106** can comprise a loop style attachment. The loop style attachment can include an adjustment mechanism connected to a top layer **104T** of the central unit and an attachment rod extending from the adjustment mechanism. In some embodiments, the attachment rod can loop around the bottom layer **104B** and standing platform **199** such that, when the attachment rod is tightened, the attachment rod can hold the top layer **104T** and bottom layer **104B** together around standing platform **199**. The attachment rod can have a fixed end configured to attach to the top layer **104T** and a free end configured to fit under the standing platform **199** of the scooter **198** and to re-engage with the adjustment mechanism. When the free end of the attachment rod is re-engaged with the adjustment mechanism, the clamp mechanism **106** can form a loop. The loop can be adjustably tightened and/or loosened. In some embodiments, the clamp mechanism **106** can be adjustable such that the loop is between about two inches in diameter and about five inches in diameter along one dimension. In some embodiments, the adjustment mechanism can have at least one tightening screw configured to selectively rotate in order to advance or retract the free end of the attachment rod through a portion of the adjustment mechanism. The selective rotation of the at least one tightening screw can adjust the overall diameter of the loop formed by the attachment rod in order to tighten or loosen the at least one clamp mechanism when the at least one clamp mechanism is engaged with the portion of the standing platform **199**.

(48) In some embodiments (not pictured), the clamp mechanism **106** can comprise a buckle attachment. A buckle can comprise a buckle head and a buckle receiver configured to receive the buckle head and lock it in place. A buckle head can connect to the top layer **104T**, and a corresponding buckle receiver can connect to the bottom layer **104B**. This configuration can be reversed such that, for example, a buckle receiver can connect to the top layer **104T**, and a corresponding buckle head can connect to the bottom layer **104B**. In some embodiments, the buckle can be vertically oriented such that the buckle head points into the buckle receiver. The buckle can enable the top layer **104T** and bottom layer **104B** to be attached to one another via vertical movement. In some embodiments, when the buckle is in a buckled position (e.g., the buckle head is locked in place within the buckle receiver), the top layer **104T** and bottom layer **104B** may be secured flush against one another. In some embodiments, the buckle can be unbuckled (e.g., the buckle head is not within the buckle receiver), and the top layer **104T** and the bottom layer **104B** can be separated.

(49) In some embodiments (not pictured), the clamp mechanism **106** can comprise a hinged attachment. A hinge can connect to a first side of the top layer **104T** and a first side of the bottom layer **104B**. The hinge can enable the top layer **104T** and bottom layer **104B** to be selectively angled away from or toward one another. In some embodiments, when the hinge is in a closed position, the top layer **104T** and bottom layer **104B** may be positioned parallel to and flush against one another. In some embodiments, the hinge can open such that a portion of the top layer **104T** and a portion of the bottom layer **104B** move/angle away from one another, creating room for the top layer **104T** and bottom layer **104B** to be placed around a standing platform **199**. Once the device **100** is positioned in the desired location, the hinge can close such that a portion of the top layer **104T** and a portion of the bottom layer **104B** are flush against one another again. This closes the top layer **104T** and bottom layer **104B** together around the standing platform **199**. In some embodiments, the hinged attachment can be used in conjunction with the buckle attachment to secure a second side of the top layer **104T** and a second side of the bottom layer **104B** together. The second sides may be different from the first sides. In some embodiments, when the buckle

attachment is buckled, the hinge attachment may be prevented from opening. This can prevent the hinge attachment from unintentionally being opened during operation of the scooter **198**.

(50) FIG. 1C depicts a lower perspective view of an exemplary balance assistance and fireproofing device compatible with a scooter, in accordance with some embodiments. As shown in FIG. 1C, beneath the standing platform **199**, a rod **108** runs underneath the central unit and is connected to two wheels. In some embodiments, the two wheels are attached to the rod such that the two wheels are situated on opposite sides of the top layer **104T** and bottom layer **104B** once attached. In some embodiments, the rod **108** is connected to the central unit via one or more brackets **110**. In some embodiments, the rod **108** is connected to the bottom layer **104B** and/or the top layer **104T** of the central unit.

(51) In some embodiments, the rod **108** can be any rod, axle, spindle, shaft, and/or similar component. In some embodiments, the rod **108** serves as an axle that is connected to two wheels, a left wheel **102L** and a right wheel **102R**. The wheels **102L** and **102R** can attach to the rod **108** using any wheel-axis attachment mechanism. The rod **108** can be a straight, linear rod that does not bend and/or twist along its length. However, in some embodiments, such as shown in FIG. 1C, the rod **108** can be a non-linear rod that bends and/or twists along its length. In some embodiments, the rod **108** can be a U-shaped rod. Depending on the size of the wheels relative to the distance between the rod **108** and the ground, the shape of the rod **108** can be adjusted. For example, the wheels **102L** and **102R** may be large such that the wheel centers are located above the bottom surface of the bottom layer **104B**. In such cases, the rod **108** can be bent upward at the ends to provide a higher attachment point (relative to the ground) at the wheels **102L** and **102R**. The rod **108** can be bent downward at the center to provide a lower attachment point when attaching to the bottom layer **104B**, which is lower (relative to the ground) than the center of the wheels **102L** and **102R**. The opposite direction of bending can occur for wheels which are small such that the wheel centers are located below the bottom surface of the bottom layer **104B**.

(52) In some embodiments, the rod **108** can be attached to the central unit at one or more fixed points. The fixed points can be locations on the bottom surface of the bottom layer **104B**, allowing the rod **108** to run underneath the central unit without intersection with the standing platform **199**. In some embodiments, the brackets **110** can be used to attach the rod **108** to the central unit at the one or more fixed points. The brackets **110** can include any suitable bracket, such as a U-shaped clamp or pipe clamp. In some embodiments, the brackets **110** can be removably attached to any portion of the central unit, such as the bottom layer **104B** or top layer. In some embodiments, the rod can be fixed at least partially within the bottom layer **104B** or top layer of the central unit, allowing the rod to run through the central unit without intersection with the standing platform **199**.

(53) Although FIG. 1C depicts the rod **108** as running underneath of the central unit and being connected to the bottom layer **104B**, it is to be understood that the rod **108** is not limited to this configuration. In some embodiments, the rod **108** can be attached to any portion of top layer **104T** and/or bottom layer **104B** of the central unit.

(54) FIG. 6 depicts a close-up view of an exemplary rod **608** of a balance assistance and fireproofing device (e.g., device **100** of FIG. 1), in accordance with some embodiments. The rod **608** is a U-shaped rod that is configured to run along a surface (e.g., an underside) of the device. The rod **608** is configured to connect one or more wheels (e.g., wheels **102L** and **102R** of FIG. 1) to the central unit. In some embodiments, the ends of the rod **608** can rotate axially due to the movement of the rod **608**, which can be undesirable due to its effect on moving the wheels up and down. To address this, the rod **608** can be fixed in place against the device so that it does not move and/or rotate axially. In some embodiments, such as shown in FIG. 6, at least a portion **608x** of the rod **608** can have a non-circular cross-section. Having such a non-circular cross-section may prevent the rod **608** from rotating axially, keeping the orientation of the rod steady during operation of the scooter. For example, the cross-section of the portion **608x** may be square or rectangular instead of circular. In some embodiments, the portion **608x** may rest flush against the bottom layer

(e.g., bottom layer **104B** of FIG. 1) and the bracket (e.g., bracket **110** of FIG. 1), preventing the rod **608** from rotating axially. The rod **608** can share any characteristics of the rod **108** of FIG. 1, and vice versa.

(55) FIG. 7 depicts a close-up view of a bracket **710** of a balance assistance and fireproofing device (e.g., device **100** of FIG. 1), in accordance with some embodiments. The bracket **710** may be configured to attach a rod to a central unit of the device. The bracket **710** may be compatible with the rod **108** of FIG. 1 and/or the rod **608** of FIG. 6. As shown in FIG. 7, the bracket **710** can be a U-shaped bracket with a groove for receiving a portion of the rod. The shape of the groove can mirror the shape of the rod such that, if the cross-section of a portion of the rod is non-circular, the bracket **710** can hold the rod stationary without allowing for axial rotation. For example, if the cross-section of the rod is square or rectangular, the sides of the groove can match the square shape of the rod, preventing the rod from rotating. In some embodiments, the bracket **710** can include one or more mechanisms for fasteners to connect the bracket **710** to the central unit. For example, the bracket **710** can include a plurality of through-holes through which a plurality of screws can extend. The screws can be tightened to hold the bracket **710** in place, attaching it to the central unit. The bracket **710** can share any characteristics of the bracket **110** of FIG. 1, and vice versa.

(56) The examples provided above describe exemplary configurations in which the top layer **104T** is a component that extends at least partly on top of the standing platform **199** and the bottom layer **104B** is a component that extends at least partly underneath the standing platform **199**. In such configurations, the top layer **104T** is positioned above the bottom layer **104B** when attached to the scooter **198**. However, it is to be understood that the terms “top” and “bottom” are merely a means of nomenclature. The terms “top” and “bottom” indicate relative positioning and do not limit the top layer **104T** to being oriented exclusively on top of the bottom layer **104B**, and vice versa. In some embodiments, the top layer **104T** and the bottom layer **104B** may be designed such that the bottom layer **104B** can be oriented on top of the top layer **104T**, and vice versa. In some embodiments, the top layer **104T** and the bottom layer **104B** may be designed such that the layers can be oriented on the sides of the standing platform **199**, and such that neither layer is above or below the other.

(57) The examples provided above describe exemplary configurations in which the device **100** and scooter **198** are distinct components, and the device **100** can be removably attached to the scooter **198**. In some embodiments, the device **100** can be integrated with the scooter **198** such that the device **100** is not removable from the scooter **198**. In such embodiments, the device **100** can be considered to be a component of the scooter **198**.

(58) In some embodiments, the scooter **198** comprises a scooter with balance assistance and fireproofing functionalities. Similar to the configurations described previously herein, the scooter **198** can comprise a scooter body comprising a standing platform (e.g., standing platform **199**) and a battery at least partially embedded within the standing platform. However, in this exemplary configuration, the scooter **198** can further include a balance assistance and fireproofing device **100** configured to be attached to the scooter body. The device **100** can include a central unit. The central unit can be attached to at least a portion of the scooter body, such as the standing platform. In some embodiments, the central unit is connected to a portion of the standing platform enclosing the battery at least partially embedded within the standing platform.

(59) In some embodiments, the central unit can comprise a top layer **104T** and a bottom layer **104B**. The layers **104T** and **104B** may be configured to be fused together such that the device **100** is integrated into the scooter **198**. For example, one or both of the layers **104T** and **104B** may be at least partially fused with the standing platform. In some embodiments, the central unit does not comprise distinct layers and can be integrally formed along with the scooter body. The central unit can share any features of the central units of the configurations described previously herein. For example, the central unit can comprise a fire-suppressant material, and/or the central unit can extend laterally outward from the standing platform.

(60) The scooter **198** can further include a rod **108** attached to the central unit and two wheels **102L** and **102R** attached to the rod such that each wheel is situated on a diametrically opposite side of the standing platform from the other wheel (e.g., on the left and on the right side of the standing platform). The rod **108** and wheels **102L** and **102R** can share any features of their counterparts in the configurations described previously herein.

(61) For the purpose of clarity and a concise description, features are described herein as part of the same or separate embodiments; however, it will be appreciated that the scope of the disclosure includes embodiments having combinations of all or some of the features described.

Additional Definitions

(62) Unless defined otherwise, all terms of art, notations and other technical and scientific terms or terminology used herein are intended to have the same meaning as is commonly understood by one of ordinary skill in the art to which the claimed subject matter pertains. In some cases, terms with commonly understood meanings are defined herein for clarity and/or for ready reference, and the inclusion of such definitions herein should not necessarily be construed to represent a substantial difference over what is generally understood in the art.

(63) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It is also to be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It is further to be understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used herein, specify the presence of stated features, integers, steps, operations, elements, components, and/or units but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, units, and/or groups thereof.

(64) The above description is presented to enable a person skilled in the art to make and use the disclosure, and is provided in the context of a particular application and its requirements. Various modifications to the preferred embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments and applications without departing from the spirit and scope of the disclosure. Thus, this disclosure is not intended to be limited to the embodiments shown, but is to be accorded the widest scope consistent with the principles and features disclosed herein.

Embodiments

(65) Disclosed herein are exemplary embodiments for the balance assistance and fireproofing device. The following description is presented to enable a person of ordinary skill in the art to make and use the various embodiments. Descriptions of specific devices, techniques, and applications are provided only as examples. Various modifications to the examples described herein will be readily apparent to those of ordinary skill in the art, and the general principles defined herein may be applied to other examples and applications without departing from the spirit and scope of the various embodiments. Thus, the various embodiments are not intended to be limited to the examples described herein and shown, but are to be accorded the scope consistent with the claims

(66) Embodiment 1. An electric scooter training wheel stability attachment for electric scooters having a main body forming a platform on which a rider stands, the main body having a top lateral end and a bottom lateral end disposed opposite the top lateral end and extending in a downward direction, the scooterizer training wheels further including two tires/wheels situated at the right and left lateral end of the rod, the stability attachment comprising: a frame having a top end and a bottom end disposed opposite, the frame being selectively mountable on the main body of the electric scooter and extends horizontally outwardly and generally transversely thereto when mounted thereon; at least both tires/wheels being situated on the rod connected to the stability attachment. the right wheel and the left wheel being situated on diametrically opposite sides of the stability attachment component which is connected to the electric scooter when the stability attachment is mounted thereon; and means for attaching the frame to the electric scooter.

- (67) Embodiment 2. A stability attachment for electric scooters as defined by embodiment 1, wherein the means for attaching the frame to the electric scooter comprises: at least one clamp mechanism affixed to the frame and situated between the top axial end and the bottom axial end of the frame, the at least one clamp mechanism being engageable with a portion of the stability attachment of the electric scooter to selectively mount the frame thereto.
- (68) Embodiment 3. A stability attachment for electric scooters as defined by embodiment 2, wherein the clamp mechanism is adjustment mechanism and an attachment rod extending therefrom, the attachment rods having a free end that may be fitted under the stability attachment of the electric scooter and engaged with the adjustment mechanism, the adjustment mechanism having at least one tightening screw that is selectively rotatable to advance and retract the free end of the attachment rod through the adjustment mechanism; and wherein the selective rotation of the tightening screw adjusts the overall diameter of the attachment rod to tighten or loosen the attachment clamp to the stability attachment of the electric scooter when the stability attachment is mounted thereon,
- (69) Embodiment 4. A stability attachment for an electric scooter as defined by embodiment 3, wherein the clamp is adjustable between about two inches in diameter and about five inches in diameter.
- (70) Embodiment 5. A stability attachment for an electric scooter as defined by embodiment 2, wherein the clamp mechanism further comprises: the top segment and a bottom segment situated opposite the top segment, the top segment and the bottom segment being attachable to and being situatable on diametrically opposite sides of the stability attachment the electric scooter when the stability attachment is mounted thereon, each of the top segment and the bottom segment having a first end and a second end disposed opposite the first end; wherein the top segment and the bottom segment are engageable with one another define the clamp mechanism.
- (71) Embodiment 6. A device comprising in combination: an electric scooter having a main body forming a platform on which a rider stands, the main body having a top lateral end and a bottom lateral end disposed opposite the first lateral end and extending in a direction between, the electric scooter vehicle further including a right and left wheel situated at opposite ends of the stability attachment; and a stability attachment for mounting to electric scooters, the stability attachment including: a frame having a top end and a bottom end disposed opposite ends, the frame being selectively mountable on the main body of the electric scooter and extending horizontally outwardly therefrom and generally transversely thereto when mounted thereon; at least a right tire and a left tire, the right tire being situated on the rod in proximity to the left tire thereof, and the right tire and the left tire being situated on diametrically opposite sides of the rod stability attachment of the electric scooter.
- (72) Embodiment 7. A balance assistance and fireproofing device configured to be attached to an electric scooter, the device comprising: a central unit comprising a top layer and a bottom layer, wherein the central unit comprises a fire-suppressant material, and wherein the central unit is configured to extend horizontally outward from a standing platform of the electric scooter when attached to the electric scooter; at least one a clamp mechanism situated between the top layer and the bottom layer of the central unit; a rod attached to the central unit at one or more fixed points; and two wheels attached to the rod such that each wheel is situated on a diametrically opposite side of the electric scooter from the other wheel.
- (73) Embodiment 8. The device of embodiment 7, wherein the at least one clamp mechanism is engageable with a portion of the standing platform of the electric scooter.
- (74) Embodiment 9. The device of embodiment 7, wherein the at least one clamp mechanism engages with the portion of the standing platform enclosing at least a portion of a battery at least partially embedded within the standing platform.
- (75) Embodiment 10. The device of embodiment 7, wherein the at least one clamp mechanism comprises a top segment and a bottom segment configured to be positioned on diametrically

opposite sides of the portion of the standing platform, wherein the top segment and the bottom segment are engageable with one another to define the at least one clamp mechanism.

(76) Embodiment 11. The device of embodiment 10, wherein the top segment comprises a groove and/or protrusion on a top layer of the central unit and the bottom segment comprises a protrusion and/or groove on a bottom layer of the central unit.

(77) Embodiment 12. The device of embodiment 10, wherein the top layer and the bottom layer are configured to slide horizontally together such that the groove and/or protrusion of the top segment engages with the protrusion and/or groove of the bottom segment to attach the central unit around a portion of the standing platform.

(78) Embodiment 13. The device of embodiment 7, wherein the at least one clamp mechanism comprises an adjustment mechanism and an attachment rod extending from the adjustment mechanism, the attachment rod having a free end configured to fit under the standing platform of the electric scooter and to engage with the adjustment mechanism, and the adjustment mechanism having at least one tightening screw configured to selectively rotate in order to advance or retract the free end of the attachment rod through a portion of the adjustment mechanism.

(79) Embodiment 14. The device of embodiment 13, wherein the selective rotation of the at least one tightening screw adjusts the overall diameter of the attachment rod in order to tighten or loosen the at least one clamp mechanism when the at least one clamp mechanism is engaged with the portion of the standing platform.

(80) Embodiment 15. The device of embodiment 13, wherein the at least one clamp mechanism is adjustable between about two inches in diameter and about five inches in diameter.

(81) Embodiment 16. The device of embodiment 7, wherein the central unit comprises a storage chamber.

(82) Embodiment 17. The device of embodiment 7, wherein the fire-suppressant material comprises a fireproof sealant comprising one or more of: caulk, foam, silicone, and putty.

(83) Embodiment 18. The device of embodiment 7, wherein the one or more fixed points are located beneath a bottom surface of the bottom layer of the central unit.

(84) Embodiment 19. The device of embodiment 7, wherein the rod is attached to the central unit at the one or more fixed points by one or more brackets.

(85) Embodiment 20. The device of embodiment 7, wherein the rod is fixed at least partially within the bottom layer of the central unit.

(86) Embodiment 21. An electric scooter with balance assistance and fireproofing, the scooter comprising: a scooter body comprising a standing platform and a battery at least partially embedded within the standing platform; and a balance assistance and fireproofing device configured to be attached to the scooter body, the device comprising: a central unit, wherein the central unit comprises a fire-suppressant material, the central unit is attached to the standing platform, and the central unit is configured to extend horizontally outward from the standing platform; at least one a clamp mechanism situated between the top layer and the bottom layer of the central unit; a rod attached to the central unit at one or more fixed points; and two wheels attached to the rod such that each wheel is situated on a diametrically opposite side of the standing platform from the other wheel.

(87) Embodiment 22. The scooter of embodiment 21, wherein the central unit is connected to a portion of the standing platform enclosing the battery at least partially embedded within the standing platform.

Claims

1. A device configured to be attached to a scooter, the device comprising: a first layer removably attached to a second layer, the first layer comprising a first groove located in a middle portion of the first layer that is configured to receive a standing platform of the scooter and at least two

second grooves located on opposite sides of the first groove of the first layer, the second grooves configured to receive corresponding protrusions of the second layer, such that the first and second layers surround at least a portion of the standing platform when attached to the scooter, wherein the first layer and/or second layer comprise a fire-suppressant material; a rod attached to the first or second layer; and two wheels attached to the rod such that the two wheels are situated on opposite sides of the attached first and second layers.

2. The device of claim 1, wherein the first layer and the second layer are configured to slide together such that the second grooves of the first layer engage with the corresponding protrusions of the second layer to surround at least a portion of the standing platform of the scooter.

3. The device of claim 1, wherein the fire-suppressant material comprises a fireproof sealant comprising one or more of: caulk, foam, silicone, and putty.

4. The device of claim 1, wherein the first or second layer comprises one or more brackets and the rod is attached to the one or more brackets.

5. The device of claim 1, wherein the rod is attached to a bottom surface of the second layer.

6. The device of claim 1, wherein the first layer and/or second layer extends laterally outward from the standing platform of the scooter when attached to the scooter.

7. The device of claim 1, wherein the standing platform comprises a battery and at least a portion of the standing platform comprising the battery is surrounded by the first and second layers.

8. A scooter system comprising: a scooter comprising a standing platform and a battery at least partially embedded within the standing platform; and a device configured to be attached to the scooter, the device comprising: a first layer removably attached to a second layer, the first layer comprising a first groove located in a middle portion of the first layer that is configured to receive the standing platform of the scooter and at least two second grooves located on opposite sides of the first groove of the first layer, the second grooves configured to receive corresponding protrusions of the second layer, such that the first and second layers surround at least a portion of the standing platform of the scooter when attached to the scooter, wherein the first layer and/or second layer comprise a fire-suppressant material; a rod attached to the first or second layer; and two wheels attached to the rod such that the two wheels are situated on opposite sides of the attached first and second layers.

9. The scooter system of claim 8, wherein the first layer and the second layer are configured to slide together such that the second grooves of the first layer engage with the corresponding protrusions of the second layer to surround at least a portion of the standing platform of the scooter.

10. The scooter system of claim 8, wherein the fire-suppressant material comprises a fireproof sealant comprising one or more of: caulk, foam, silicone, and putty.

11. The scooter system of claim 8, wherein the rod is attached to a bottom surface of the second layer.

12. The scooter system of claim 8, wherein the first layer and/or second layer extends laterally outward from the standing platform of the scooter when attached to the scooter.

13. The scooter system of claim 8, wherein the standing platform comprises a battery and at least a portion of the standing platform comprising the battery is surrounded by the first and second layers.
